

Machine Learning Basics

A Simple Reference Document for a Retrieval-Augmented Generation (RAG) Project

1. What is Machine Learning?

Machine Learning (ML) is a branch of artificial intelligence that allows computers to learn patterns from data and use those patterns to make predictions or decisions. Instead of explicitly programming every rule, we provide examples and allow a model to learn a relationship between inputs and outputs.

For example, a model can be trained using historical house data containing features such as area, number of rooms, and location. After learning from the examples, the model can estimate the price of a new house.

2. Types of Machine Learning

Machine learning is commonly divided into supervised learning, unsupervised learning, and reinforcement learning. The main difference is the type of information available during learning.

2.1 Supervised Learning

In supervised learning, the training data contains both input features and known target labels. The model learns a mapping from the inputs to the desired output. Common supervised learning tasks include classification and regression.

Classification predicts a category. For example, an email classifier may predict whether an email is spam or not spam. Regression predicts a numerical value, such as predicting the price of a house.

2.2 Unsupervised Learning

In unsupervised learning, the data does not contain target labels. The algorithm attempts to discover useful patterns or structures within the data. Clustering is a common example. A clustering algorithm may group customers according to similarities in their purchasing behavior.

2.3 Reinforcement Learning

Reinforcement learning is based on interaction between an agent and an environment. The agent takes actions and receives rewards or penalties. Over time, it learns a strategy, called a policy, that aims to maximize the total reward.

3. Training and Testing

A machine learning dataset is commonly divided into training and testing sets. The training set is used to learn model parameters, while the test set is used to evaluate how well the trained model performs on previously unseen examples.

A model that performs well on training data but poorly on unseen data may have memorized the training examples instead of learning patterns that generalize.

4. Overfitting and Underfitting

Overfitting occurs when a model learns the training data too closely, including noise or accidental patterns. Such a model may have high training performance but poor performance on new data.

Underfitting occurs when a model is too simple to capture the important patterns in the data. It may perform poorly on both the training and testing sets.

A good machine learning model attempts to find a balance in which it learns useful patterns while still generalizing well to unseen examples.

5. Common Machine Learning Algorithms

Different machine learning problems require different algorithms. Linear regression is commonly used for predicting continuous numerical values. Logistic regression can be used for classification problems. Decision trees make predictions using a sequence of feature-based decisions.

Support Vector Machines attempt to find a decision boundary that separates classes. K-Nearest Neighbors predicts a new example based on nearby examples. Neural networks use layers of interconnected computational units and are especially useful for complex tasks involving images, text, audio, and other high-dimensional data.

6. Features and Labels

Features are the input variables used by a machine learning model. A label, also called a target, is the value the model is trying to predict in supervised learning.

For a house-price prediction problem, house size, number of bedrooms, and location can be features, while the house price is the label. Choosing useful features can significantly affect model performance.

7. Model Evaluation

Model evaluation measures how well a machine learning model performs. The appropriate metric depends on the problem. Accuracy measures the proportion of correct predictions in a classification task. Precision measures how many predicted positive examples are actually positive, while recall measures how many actual positive examples were found.

For regression problems, metrics such as Mean Squared Error (MSE) and Mean Absolute Error (MAE) can be used. Selecting the correct evaluation metric is important because a single metric may not fully describe model behavior.

8. Neural Networks

A neural network is a machine learning model made of layers of interconnected units. The input layer receives features, hidden layers transform the information, and the output layer produces a prediction.

During training, the network calculates a loss that measures the difference between its prediction and the target. An optimization algorithm then updates the model's parameters to reduce the loss. Backpropagation is used to calculate gradients of the loss with respect to model parameters.

9. Deep Learning

Deep learning is a subfield of machine learning that uses neural networks with multiple layers. Deep learning models can automatically learn increasingly complex representations from raw data.

Convolutional Neural Networks (CNNs) are commonly used for image-related tasks. Recurrent Neural Networks (RNNs) were designed for sequential data, while Transformer models use attention mechanisms and are widely used for natural language processing.

10. Key Terms

Dataset: A collection of examples used for analysis or machine learning.

Feature: An input variable provided to a machine learning model.

Label: The target value a supervised learning model attempts to predict.

Training: The process of learning model parameters from data.

Inference: Using a trained model to make predictions on new data.

Epoch: One complete pass through the training dataset.

Loss: A value that represents how different a model's prediction is from the target.

Gradient: A value that indicates how a model parameter should change to reduce loss.

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